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DEPARTMENT OF DEFENSE
DESIGN CRITERIA STANDARD

DEFINITIONS OF AND BASIC REQUIREMENTS FOR ENCLOSURES FOR ELECTRIC AND ELECTRONIC EQUIPMENT



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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. An enclosure is a mechanical item that wholly or partly surrounds some electrical or electronic item or group of items and is an integral part of the equipment. It is used herein in a general sense to include such items as cabinet and housing which are defined in the Federal Item Identification Guide for Supply Cataloging. It may perform any or all of the following functions:

a. Provide physical protection to the enclosed items.
b. Provide physical protection from items within the enclosure to personnel or objects outside the enclosure.
c. Provide means for ventilating items within the enclosure.
d. Provide means for cooling items within the enclosure.
e. Provide external facilities for mounting on, or securing to, a structure, such as a wall or bulkhead.
f. Provide interior support for subassemblies, parts, and similar items.
g. Provide reduction of radiated audible noise or radio interference.

This standard defines the enclosure by the environmental protection that it provides to its contents. In some cases, this standard covers physical protection to the equipment or to personnel. Specific requirements for ventilation, cooling, mounting, support, and reduction of audible noise or radio interference should be covered by equipment specifications.

3. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, D.C. 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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CONTENTS

PARAGRAPH PAGE
FOREWORD ii
CONTENTS iii
1. SCOPE 1
1.1 Scope 1
2. APPLICABLE DOCUMENTS. 1
2.1 General 1
2.2 Government documents 1
2.2.1 Specifications, standards, and handbooks 1
2.2.2 Other Government documents, drawings, and publications 1
2.3 Non-Government publications 1
2.4 Order of precedence 2
3. DEFINITIONS 2
3.1 Definitions for degrees of enclosures 2
3.2 Supplementary definitions for enclosure identification by cooling method 3
4. GENERAL REQUIREMENTS 4
4.1 Use of definitions for parts 4
4.2 Basic requirements and tests 4
4.3 Design requirements for hazardous atmospheres. 6
4.3.1 Dust-ignition proof 6
4.3.2 Explosionproof 6
4.3.3 Protection against contact with external fans 6
4.4 Construction 6
4.5 Recycled, recovered, environmentally preferable, or biobased materials 6
5. DETAILED REQUIREMENTS 7
5.1 Testing of parts 7
5.2 Test procedures 7
5.3 Airtight 7
5.4 Drip-proof (15 degrees) and drip-proof (45 degrees) 7
5.5 Dust-ignition proof 8
5.6 Dustproof 8
5.7 Dusttight 8
5.8 Explosionproof 8
5.9 Hermetic 8
5.10 Splashproof 8
5.11 Spraytight 10
5.12 Submersible and open-submersible 11
5.12.1 Submersible and open-submersible electrical test 11
5.13 Totally enclosed 11
5.14 Watertight 11
6. NOTES 11
6.1 Intended use 11
6.2 Acquisition requirements 11
6.3 Other rating documents used for degrees of protection 11
6.4 Subject term (key word) listing 13
6.5 Changes from previous issue 13

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LIST OF TABLES

TABLE PAGE
TABLE I. Definitions. 2
TABLE II. Design requirements and tests. 4
TABLE III. Water repellent tests using hose. 9
TABLE IV. Submergence tests. 10
TABLE V. NEMA and IEC rating comparison. 11

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1. SCOPE

1.1 Scope. This standard establishes definitions and basic requirements for enclosures for environmental protection of electrical and electronic equipment.

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-6983506 - Electrical Enclosure, Composite Installation and Details

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@us.navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60529 - Degrees of Protection Provided by Enclosures

(Copies of this document are available online at www.iec.ch.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 70 - National Electric Code

(Copies of this document are available online at www.nfpa.org.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 50 - Standard for Enclosures for Electrical Equipment, Environmental Considerations

UL 1203 - Standard for ExplosionProof and Dust-IgnitionProof Electrical Equipment for Use in Hazardous (Classified) Locations

(Copies of this document are available online at www.ul.com.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Definitions for degrees of enclosures. Degrees of enclosures for environmental protection of electrical and electronic equipment are defined in table I. Enclosures are defined by either of two general methods that are consistent with industry practice:

a. Enclosures defined solely in terms of protection provided by the enclosure itself; the function of the enclosure being to exclude the undesirable or harmful element. All degrees of enclosure having the suffix "tight" fall in this category.

b. Enclosures defined in terms of specified environmental conditions under which the enclosed equipment (equipment with its enclosure) will operate satisfactorily. The ability of such enclosed equipment to perform under the specified conditions is not necessarily wholly dependent on the enclosure itself but may also depend upon the characteristics of the equipment. All degrees of enclosure having the suffix "proof" fall in this category.

TABLE I. Definitions.

Degree of Enclosure	Definition
Airtight 1/	Enclosure prevents transmission of air between interior and exterior.
Drip-proof (15 degrees) 2/ Drip-proof (15 degrees) 2/, protected 3/ Drip-proof (45 degrees) Drip-proof (45 degrees), protected	Enclosed equipment operates satisfactorily in presence of: (1) Drops of liquid or solid particles falling at angles up to 15 (or 45) degrees from the vertical with enclosed equipment mounted in its normal position. (2) Drops of liquid or solid particles falling vertically with enclosed equipment inclined at angles up to 15 (or 45) degrees from its normal mounting position.
Dust-ignition proof	Enclosed equipment operates satisfactorily, without hazard, in presence of ignitable amounts of a specified dust. The enclosure is designed and constructed to: (1) Exclude ignitable amounts of dust. (2) Exclude amounts of dust that might affect performance or rating. (3) Prevent ignition of exterior accumulations or atmospheric suspensions of a specific dust on or in the vicinity of the enclosure by arcs, sparks, or heats generated or liberated within the enclosure.
Dustproof	Enclosed equipment operates satisfactorily in presence of dust.
Dusttight	Enclosure prevents entry of dust.
Explosionproof	Enclosed equipment operates safely in presence of any concentration of a specified gas or vapor. The enclosure is designed and constructed to withstand without distortion or other significant damage explosions of a specified gas or vapor, which may occur within it and prevent ignition of the specified gas or vapor surrounding the enclosure by sparks, flashes, or explosion of the gas or vapor within.
Hermetic 1/	Enclosure is permanently sealed to prevent transmission of air, gas, or moisture.
Open Open, protected 3/	Enclosure provides no environmental protection and permits free transmission of air.

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TABLE I. Definitions – Continued.

Degree of Enclosure	Definition
Splashproof Splashproof, protected [3/]	Enclosed equipment, when mounted in its normal manner, operates satisfactorily in presence of liquid (coarse spray) or solid particle directed at it or its mounting surface.
Spraytight [4/]	Enclosure prevents entry of liquid striking it from any direction.
Submersible (15 feet) Submersible (50 feet) Submersible (150 feet) Submersible (300 feet) Submersible (1,600 feet) Open submersible (15 feet) Open submersible (50 feet) Open submersible (150 feet) Open submersible (300 feet) Open submersible (1,600 feet)	Enclosed equipment operates satisfactorily when submerged in water up to specified depths.
Totally enclosed	Enclosure prevents circulation of air between the exterior and interior but does not necessarily prevent leakage of air.
Watertight	Enclosure prevents leakage of water striking or covering enclosure.

NOTES:

[1/] Not applicable for rotating electrical equipment such as motors and generators.

[2/] Drip-proof, if used without a specified angle, refers to drip-proof (15 degrees).

[3/] Protected, when used as part of an enclosure designation, indicates that ventilation or other enclosure openings, where permitted, are limited in size to prevent accidental contact with live electrical or mechanically moving parts. To afford this protection, openings shall be of such size and shape as to prevent passage of a 0.50-inch diameter rod, except where the distance from the exterior of the opening to the live or moving parts is more than 4 inches, in which case the openings shall be for such size and shape as to prevent passage of a 0.85-inch diameter rod.

[4/] Applicable only to rotating electrical equipment such as motors and generators.

3.2 Supplementary definitions for enclosure identification by cooling method. The degrees of enclosure listed in table I are commonly used without supplementary identification, but it is also a frequent practice to further identify enclosed equipment by the method of cooling. Standard terms used for the methods of cooling and their purpose are listed below. When these terms are used with reference to enclosures, they should be in combination with and following the degree of environmental protection (e.g., totally enclosed, fan-cooled). This standard does not contain separate design requirements or testing requirements for the methods of cooling. The requirements in this document are applicable to each degree of enclosure identified in table I, regardless of method of cooling specified (e.g., requirements specified herein for explosionproof, spraytight, and totally enclosed equipment also apply respectively to explosionproof [fan-cooled], spraytight [fan-cooled], and totally enclosed [fan-cooled] equipment).

a. Air-to-air cooled. Equipment is cooled by circulating the internal air through a heat exchanger which, in turn, is cooled by circulating external air. An air-to-air exchanger, a fan, or fans for circulation of internal air and a fan for circulation of external air are provided.

b. Fan-cooled.Used for rotating equipment only. Equipment is cooled by air flow over the enclosed machine. Air flow is produced by a fan (or fans) attached to the equipment shaft. Protective covers are provided for fans (see 4.3.3).

c. Ventilated. Equipment is cooled by air passing through the enclosure. Specific terms used to describe ventilated equipment are:

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(1) Natural-convection ventilated. Air circulation through the enclosure by natural convection.

(2) Self-ventilated. Generally used for rotating machines such as motors and generators. Air is circulated through the enclosure by a means integral with the equipment.

(3) Forced-air-ventilated, Forced-draft-ventilated. Generally used for other than rotating equipment. Air is circulated through the enclosure by a fan or blower located within the enclosure.

(4) Separately ventilated. Equipment is intended for cooling by separately supplied forced air from a source other than the atmosphere surrounding the enclosure. Enclosure is provided with inlet and outlet air duct connections.

d. Non-ventilated. Generally used for rotating equipment such as motors and generators. Enclosure has no ventilation openings or provision for air flow other than natural air convection over the enclosed equipment. Rotating equipment may be provided with an internal fan (or fans) attached to the shaft to recirculate air within the enclosure.

e. Water-cooled. Equipment is cooled by circulating water. The circulating water or the water conductors come into direct contact with parts to be cooled.

f. Water-air-cooled. Equipment is cooled by circulating air which, in turn, is cooled by circulating water. A water-cooled heat exchanger and a fan (or fans) for circulating the air are provided.

4. GENERAL REQUIREMENTS

4.1 Use of definitions for parts. The definitions of this standard may be applied to parts (e.g., fuse holders) designed for mounting in, and projecting through, an equipment enclosure wall or panel. A part so defined is intended for use in a similarly defined enclosure (e.g., a watertight fuse holder in a watertight enclosure) and when properly installed shall prevent the entry of the undesirable or harmful environmental element into the enclosure and the enclosed portions of the part.

4.2 Basic requirements and tests. The basic design requirements and tests applicable to a particular degree of enclosure shall be as specified in table II and section 5.

TABLE II. Design requirements and tests.

Degree of Enclosure	Enclosure Design Requirements	Test Procedure
Airtight [1/]	Designed for internal and external pressure differences up to 10 psi [2/]. Pressure change in a 24-hour period shall not exceed 6 percent of the original pressure difference.	5.3
Dripproof (15 degrees) [3/] Dripproof (15 degrees) [3/], protected [4/] Dripproof (45 degrees) Dripproof (45 degrees), protected [4/]	Designed so that drops of water or solid particles falling under defined conditions have no harmful effect on equipment operation. Ventilation openings in bottom of enclosure and louvers in sides of enclosure are permitted.	5.4
Dust-ignition proof	No ventilation or other openings are permitted. External temperature of enclosed equipment shall be such that excessive dehydration or gradual carbonization of organic dust deposits will not occur. Special requirements as specified shall apply (see 4.3.1 and 6.2).	5.5
Dustproof	Designed so that dust will not interfere with equipment operation.	5.6
Dusttight	Designed to exclude dust. No ventilation or other openings are permitted.	5.7
Explosionproof	No ventilation or other openings are permitted. External temperature of enclosed equipment shall be such that a surrounding explosive atmosphere will not be ignited. Special requirements as specified shall apply (see 4.3.2 and 6.2).	5.8

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TABLE II. Design requirements and tests – Continued.

Degree of Enclosure	Enclosure Design Requirements	Test Procedure
Hermetic [1/]	Enclosure shall be rigid, nonporous material such as metal, glass, or ceramic sealed by a fusion process, such as soldering or welding.	5.9
Open Open, protected [4/]	As a minimum, an open enclosure shall be a structural framework surrounding the equipment and offering impediment only to gross accidental contact by personnel or equipment. An open protected enclosure is intended to provide full protection against accidental contact. Open and open protected enclosures are intended only for equipment that is not subjected to environmental conditions that are harmful to the equipment.	None
Splashproof Splashproof, protected [4/]	Water (coarse spray) or solid particles directed at the enclosed equipment or its mounting surface shall have no harmful effect on equipment operation.	5.10
Spraytight [5/]	Designed so that water directed at exterior of enclosure at any angle will not enter (except for insignificant amounts along shaft). No ventilation or other openings are permitted. Gaskets are required for covers and doors. Gaskets shall be sufficient to meet the requirements of water exposure (i.e., equipment located out-of-doors or above main deck on ships.)	5.11
Submersible (15 feet) Submersible (50 feet) Submersible (150 feet) Submersible (300 feet) Submersible (1,600 feet)	Unless otherwise specified (see 6.2), submersible enclosures shall prevent leakage of water during submergence at specified depths in degree of enclosure column.	5.12
Open-submersible (15 feet) Open-submersible (50 feet) Open-submersible (150 feet) Open-submersible (300 feet) Open-submersible (1,600 feet)	Open-submersible enclosures permit entry of the liquid in which the enclosed equipment is designed to operate.	5.12
Totally enclosed	Enclosure shall completely enclose equipment. No ventilation or other openings are permitted. Enclosures shall protect against dust, dirt, steam, corrosive vapors, and water vapor in nonhazardous environments. The complete exclusion of the aforementioned conditions is not required.	5.13

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TABLE II. Design requirements and tests – Continued.

Degree of Enclosure	Enclosure Design Requirements	Test Procedure
Watertight	Water directed at enclosure from any angle or water covering enclosure shall not enter (except for insignificant amounts along shaft of rotating equipment). No ventilation or other openings are permitted. Gaskets are required for covers and doors. Watertight enclosures typically used for weather exposure and transient submergence conditions, such as found in ship's deck applications.	5.14

NOTES:

1/ Not applicable for rotating electrical equipment such as motors and generators.

2/ Pressure per pounds per square inch

3/ Drip-proof, if used without a specified angle, refers to drip-proof (15 degrees).

4/ Protected, when used as part of an enclosure designation, indicates that ventilation or other enclosure openings, where permitted, are limited in size to prevent accidental contact with live electrical or mechanically moving parts. To afford this protection, openings shall be of such size and shape as to prevent passage of a 0.50-inch diameter rod, except where the distance from the exterior of the opening to the live or moving parts is more than 4 inches, in which case the openings shall be for such size and shape as to prevent passage of a 0.75-inch diameter rod.

5/ Applicable only to rotating electrical equipment such as motors and generators.

4.3 Design requirements for hazardous atmospheres.

4.3.1 Dust-ignition proof. The type of hazardous dust involved shall be as specified (see 6.2). The requirements of NFPA 70 shall apply when specified (see 6.2).

a. NFPA 70.

b. MIL-STD-810.

c. Special Naval Shipboard requirements in accordance with MIL-DTL-2036.

d. UL 1203

4.3.2 Explosionproof. The specific hazardous gas or vapor involved shall be as specified (see 6.2). In addition, the following requirements apply when specified (see 6.2):

a. NFPA 70.

b. MIL-STD-810.

c. Special Naval Shipboard requirements in accordance with MIL-DTL-2036.

d. UL 1203.

4.3.3 Protection against contact with external fans. Fans that are external to the basic equipment enclosure (e.g., totally enclosed, fan-cooled motor) shall have protective covers, grills, or screens to prevent accidental contact with the moving parts. Openings in the covers, grills, or screens shall be of such size that a 0.50-inch diameter rod will not pass through. On spraytight fan-cooled and watertight fan-cooled machines, drain holes shall be provided to prevent accumulation of water in the fan housing.

4.4 Construction. Enclosures (including protective covers, grills, and screens) shall be of rigid construction and in accordance with MIL-DTL-2036. Composite enclosures shall be fabricated in accordance with NAVSEA Drawing 803-6983506.

4.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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5. DETAILED REQUIREMENTS

5.1 Testing of parts. Unless otherwise specified (see 6.2), testing of parts shall be conducted to the extent required by the parts specification. Where parts specifications do not cover test conditions and procedures, the test conditions and procedures of this standard apply to the part when mounted in a test enclosure that, except for the openings for part mounting, is known to prevent the entry of the undesirable or harmful environmental element (water, dust, etc.,). In the case of drip-proof parts, the following accelerated test shall apply instead of the drip test for enclosures described in 5.4.b:

a. Drip source: Sprinkler discharging water vertically downward at a minimum rate of 2 gallons per minute.

b. Sprinkling pattern: Circular and pattern not more than 12 inches in diameter, 3 feet below sprinkler.

c. Sprinkling distribution: Approximately uniform.

d. Distance from sprinkler to part: 3 feet below.

e. Orientation of part: Mounting surface inclined 15 degrees from vertical maximum exposure of part at that angle.

f. Total test time: 5 minutes.

5.2 Test procedures. The following test procedures are applicable to specific degrees of enclosure for environmental protection. Unless otherwise specified (see 6.2), tests shall be conducted to the extent specified by the equipment specification or as otherwise required and when so specified shall form a part of the inspection procedure for the equipment. Tests apply to the enclosed equipment (enclosure with contents) except that dusttight and watertight enclosures for portable equipment may be tested without the contents, provided that the contents are not to be attached to the enclosure. Failure to meet the specified requirements, including the di-electric and insulation resistance requirements of the equipment specification following enclosure tests described herein shall be cause for rejection.

5.3 Airtight. Tests for airtightness shall be conducted with both of the following initial conditions.

a. Internal air pressure 10 psi above external pressure.

b. Internal air pressure 10 psi below external air pressure.

Total time of each test shall be 24 hours minimum. The equipment operating sequence shall be as follows: ON— 8 hours, OFF—4 hours minimum; ON—8 hours, OFF—4 hours minimum. A change in pressure difference (internal minus external or external minus internal pressure) of more than 0.6 pound per square inch gauge (psig) during either test, after corrections for changes in barometric pressure and temperature, shall be cause for rejection.

5.4 Drip-proof (15 degrees) and drip-proof (45 degrees). Visual examination or actual drip test may be used to determine the ability of the enclosed equipment to operate satisfactorily under the defined conditions. Unless otherwise specified (see 6.2), either method may be used. Visual examination is intended to determine the ability of the enclosure to exclude falling drops of liquid or solid particles. Such exclusion is not necessarily required of the enclosure (see table II). If the enclosed is of such design that falling drops of liquid or solid particles appear to be able to enter, the ability of the enclosed equipment to operate satisfactorily under the defined conditions can be determined only by further examination of equipment within the enclosure or by actual drip test.

a. Visual examination. Enclosed equipment shall be visually examined for the presence of the following features and characteristics when in its normal position or inclined in any direction at angles not exceeding the specified 15 degrees or 45 degrees.

(1) There are no openings directly exposed to falling drops of water or solid particles. Openings where provided on exposed surfaces shall be protected by louvers or suitably covered.

(2) All removable access plates or covers on enclosure surfaces exposed to falling drops of water shall have gaskets. Doors and door openings on exposed enclosure surfaces shall be of such design as to prevent the entry of falling drops of water.

(3) There shall be no paths on the surface of the enclosure that drops of liquid or solid particles will follow and run into the enclosure.

b. Drip test. The enclosed equipment shall be placed below a device that drips or sprinkles water over the entire enclosed equipment. (Examples of acceptable test devices are the test equipment described in IEC 60529 and MIL-STD-810, method 506 test to evaluate materials likely to be exposed to rain, water spray, and dripping water.) The minimum amount of water shall be 1,000 cubic inches per square foot of area covered by the drip or sprinkle in 60 minutes. The dripping or sprinkle rate and distribution shall be approximately uniform. The enclosed equipment

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shall be oriented in five different positions during the test: normal position and inclined at the maximum angle (15 degrees or 45 degrees) forward, backward, and to each side. The minimum time in each position shall be 12 minutes: 8 minutes operating and 4 minutes OFF. Failure of the equipment to operate satisfactorily or accumulation of water within the enclosure shall be cause for rejection.

5.5 Dust-ignition proof. Unless otherwise specified (see 6.2), tests shall be in accordance with UL 1203for Class II, Division 1, Group E, F, G specific combustible dusts involved. .

5.6 Dustproof. Tests shall be in accordance with method 510 procedure I-blowing dust test of MIL-STD-810, except that the test at an ambient temperature of 145 °F should be omitted for equipment not designed to operate at that temperature. The equipment shall be operated during the test. Failure of the equipment to operate satisfactorily or accumulation of dust within the enclosure that would lead to eventual unsatisfactory operation shall be cause for rejection.

5.7 Dusttight. Testing shall be in accordance with method 510 procedure I-blowing dust test of MIL-STD-810, except that the test at an ambient temperature of 145 °F should be omitted for equipment not designed to operate at that temperature. Entry of any dust, revealed by subsequent disassembly and examination, shall be cause for rejection.

5.8 Explosionproof. Testing shall be in accordance with the applicable enclosure or environmental testing specification (see 4.3.2). Unless otherwise specified (see 6.2), tests shall be in accordance with UL 1203 for Class I, Division I, Group A, B, C, D specific hazardous gas or vapor involved.

5.9 Hermetic. Unless otherwise specified (see 6.2), the test procedure, sensitivity of the leakage detector, and basis for rejection of the enclosed equipment shall be as required by the equipment specification. Testing shall include filling the enclosure with helium and using a helium mass spectrometer to detect leakage.

5.10 Splashproof. Testing in accordance with table III shall be conducted with the enclosed equipment placed or mounted in a position typical of that for which it is designed. The surface upon which the equipment is placed or mounted (supporting surface) shall extend at least 3 feet beyond the equipment on all sides so that splashing may be produced by directing the water stream on that surface. The water stream shall be played upon both the equipment and supporting surface from all directions on the equipment side of the surface. Equipment shall be operated during the test. Failure of the equipment to operate satisfactorily or accumulation of water within the enclosure shall be cause for rejection.

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TABLE III. Water repellent tests using hose.

Enclosure	Rate of Water Flow (gal/min)	Form of Water Flow	Nozzle Type	Water Head (feet) [1/]	Approx. Distance from Nozzle to Enclosure (feet)	Time of Test (min)	Test Details [1/, 2/]
Splashproof	15	Coarse spray (large drops)	As required	10 [3/]	6	5	Equipment shall be operating throughout test. Water stream shall be directed at all exposed enclosure surfaces and at surface at which enclosed equipment is mounted.
Spraytight	65	Solid stream	Solid stream nozzle, 1-inch diameter tip opening	As required	10	5	Rotating equipment (motors, generators, etc.) shall be operating during the first ⅓ of the test, at standstill during the second ⅓ of the test, and operating during the final ⅓ of the test. Non-rotating equipment need not be operated during the test unless movable parts, such as switches and solenoids, penetrate the enclosure. Where movable parts penetrate the enclosure, each movable part shall be operated at least three times during the test. Water stream shall be directed at all surfaces of enclosed equipment. Enclosed equipment shall be inclined for part of the test if necessary to expose surfaces such as the underside which would not otherwise be accessible to the water stream.
Watertight (water repellent test optional see 5.14)						60	

NOTES:

1/ In testing spraytight and watertight equipment, caution shall be observed during the tests and subsequent inspection to avoid conditions that might produce condensation that could be confused with leakage.

2/ Test conditions:

a. Type of water – either fresh or salt.

b. Ambient (air) temperature – between 40 and 100 °F.

c. Water temperature – not to exceed air temperature.

3/ Water head – If the test stream is directed vertically (straight upward) for determination of required head, the stream shall rise to a height of not less than 10 feet.

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5.11 Spraytight. Testing in accordance with table III shall be conducted. At the option of the contractor, however, the submergence test of 5.12, outlined in table IV, may be substituted. The submergence test shall not be used for fan-cooled machines unless the design is such that the fan can be removed for the test and replaced afterward without difficulty. Entry of water during test by either method shall be cause for rejection, except the leakage of water less than 0.25 cubic inch per inch of shaft diameter into the bearing housing is permissible.

TABLE IV. Submergence tests.

Enclosure	Submergence Depth [1/] (feet) or Equivalent Pressure	Time of Test (minimum)	Test Details [2/, 3/, 4/]
Spraytight (submergence test optional, see 5.11)	Covered	5 minutes	Rotating equipment (motors, generators, etc.) shall be operating during the first ⅓ of the test, at standstill during the second ⅓ of the test, and operating during the final ⅓ of the test. Non-rotating equipment need not be operated during the test unless movable parts such as switches and solenoids penetrate the enclosure. Where movable parts penetrate the enclosure, each movable part shall be operated at least three times during the test.
Watertight	3 feet (1.3 psi)	1 hour	
Submersible (15 feet) Open-submersible (15 feet)	15 feet (6.5 psi)	24 hours [5/]	Equipment shall be operating during the first ⅓ of the test, at standstill during the second ⅓ of the test, and operating during the final ⅓ of the test.
Submersible (50 feet) Open-submersible (50 feet)	50 feet (21.7 psi)		
Submersible (150 feet) Open-submersible (150 feet)	150 feet (65.1 psi)		
Submersible (300 feet) Open-submersible (300 feet)	300 feet (130 psi)		
Submersible (1,600 feet) Open-submersible (1,600 feet)	1,600 feet (694 psi)		

NOTES:

1/ For convenience in conducting tests on water, submersible and open-submersible enclosed equipment, external pressure, internal vacuum, or combination thereof may be employed to create a pressure difference equivalent to the submergence depth or pressure specified. In such cases, the enclosed equipment shall be covered with a minimum of 3 inches of water.

2/ Tests may be conducted in either fresh water or salt water, except that for tests of open-submersible equipment, the water shall be that in which the equipment is designed to operate. Water temperature shall be (73±18 °F).

3/ Before conducting tests, doors and covers on enclosures shall be removed and replaced three times.

4/ Caution shall be observed during tests and subsequent inspection to avoid conditions that might produce condensation which could be confused with leakage.

5/ The 24-hour submergence test shall be considered as a design test for proving the ability of a particular equipment design to operate satisfactorily under the specified conditions. For production (quality assurance) testing, once the design has been proven, the minimum time of test shall be 1 hour.

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5.12 Submersible and open-submersible. Equipment shall be submergence tested in accordance with table IV. For submersible, the test item shall be visually inspected for evidence of water penetration. For enclosures of submersible equipment, leakage of water into any part of the enclosure, including terminal boxes, as revealed by subsequent disassembly and examination shall be cause for rejection. In addition, the enclosure, as a result of the test, shall not exhibit any structural weaknesses that could be detrimental to equipment performance.

5.12.1 Submersible and open-submersible electrical test. When parts having an internal function, such as switches or connectors, are tested to submergence requirements of this standard, a saturated saltwater solution may be used as the submergence medium in lieu of freshwater or seawater to facilitate electrical testing to detect leakage. Appropriate electrical tests shall be performed before and after submergence exposure; changes in electrical parameters measured, such as insulation resistance, contact resistance, etc., shall be evidence of enclosure failure.

5.13 Totally enclosed. The enclosure shall be visually examined for absence of openings. Joints need not be sealed or provided with gaskets; however, joined parts shall fit as tightly as expected with good manufacturing practice and workmanship.

5.14 Watertight. Testing shall be in accordance with table IV. Unless otherwise specified (see 6.2), the water repellent test of table III may be substituted in the following cases:

a. For convenience in testing equipment having one or more dimensions (length, width, or height) exceeding 5 feet.

b. For testing motors, generators, and similar rotating equipment (Note: Fans of fan-cooled watertight motors must be removed if submergence testing is used).

Entry of water during test by either method shall be cause for rejection except that for rotating equipment, water leakage of less than 0.25 cubic inch per inch of shaft diameter into any bearing housing is permissible.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard covers the basic requirements applicable to enclosures intended for naval shipboard use to ensure environmental protection of electric and electronic equipment in shipboard applications. A lesser degree of enclosure may be satisfactory for a particular application; however, such determination will be made, or approved, by the contracting activity responsible for establishing technical requirements.

6.2 Acquisition requirements. Acquisition documents should specify the following.

a. Title, number, and date of this standard.

b. Requirements for degree of enclosure(s) being procured (see 4.2).

c. When submersible enclosures are required to prevent leakage of water during submergence at depths not exceeding 1,600 feet (see table II).

d. Design requirements for hazardous atmospheres if applicable (see table II, 4.3.1, and 4.3.2).

e. When the requirements of NFPA 70, MIL-STD-810, MIL-DTL-2036, or UL 1203 apply (see 4.3.1 and 4.3.2).

f. The extent of tests to be conducted if other than those specified (see 5.2, 5.4, 5.5, 5.8, 5.9, and 5.14).

6.3 Other rating documents used for degrees of protection. This section is intended to be used as guidance only and may be used when selecting commercial enclosures that will support the degrees of protection the enclosures provide for different environments; however, all commercial NEMA and IEC rated enclosures are to be verified to meet the requirements in this document. Table V provides a list of NEMA 250 and IEC 60529 ratings that are similar to the degrees of protection identified in MIL-STD-108. If a NEMA 250 or IEC 60529 rated enclosure is selected, it does not relieve or replace the requirements of this document. Caution should be exercised when selecting the degree of protection for a particular environment, since the IEC rating does not require a corrosion test, gasket aging test, dust test or an external icing test.

TABLE V. NEMA and IEC rating comparison.

MIL-STD-108	NEMA Enclosure Type 1/	IEC 2/
Airtight	4 and 4X 6 and 6P	IP56/IP66 IP67

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TABLE V. NEMA and IEC rating comparison – Continued.

MIL-STD-108	NEMA Enclosure Type [1/]	IEC [2/]
Drip-proof 15 degrees	2 [3/] 3R and 3RX 3, 3S, and 3SX	IP11 IP14 IP54
Drip-proof 15 degrees, protected	4 and 4X 6 and 6P 12 [3/] and 12K [3/] 13 [3/]	IP56/IP66 IP67 IP52 IP54
Drip-proof 45 degrees	3R and 3RX 3, 3S, and 3SX	IP14 IP54
Drip-proof 45 degrees, protected	4 and 4X 6 and 6P 12 [3/] and 12K [3/] 13 [3/]	IP56/IP66 IP67 IP52 IP54
Dust-ignition proof	9 [3/, 4/]	IP66
Dustproof	3, 3S, and 3SX 4 and 4X 6 and 6P	IP54 IP56 IP67
Dust-tight	5 [3/] 3 and 3S 6 and 6P 12 [3/] 13 [3/]	IP52 IP54 IP67 IP52 IP54
Explosionproof	7 [3/, 5/], 8 [5/]	IP55
Hermetic	4 through 6P	IP66/IP67/IP68
Open	1 [3/]	IP10
Open, protected	1 [3/]	IP20 IP30 IP40
Splashproof	3, 3S, and 3SX	IP54
Splashproof, protected	4 through 13 [3/]	IP54/IP56
Spraytight	3, 3S, and 3SX 3X	IP54 IP55
Submersible (15 foot)	6 [6/] and 6P [7/]	IP67 and IP68
Submersible (50 foot) Submersible (150 foot) Submersible (300 foot) Submersible (1,600 foot)	6P [7/]	IP68

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TABLE V. NEMA and IEC rating comparison – Continued.

MIL-STD-108	NEMA Enclosure Type 1/	IEC 2/
Open-submersible (15 foot)	6 6/ and 6P 7/	IP67 and IP68
Open-submersible (50 foot) Open-submersible (150 foot)	6P 7/	IP68
Open-submersible (300 foot) Open-submersible (1,600 foot)		
Totally Enclosed	3, 3S, 3SX, 5 3/,12 3/, 12K 3/, 13 3/	IP52/IP54
Watertight	4 through 6P	IP66

NOTES:

1/ NEMA 250 used for ingress protection of enclosures for electrical equipment rated up to 1,000 V or less, used exclusively for enclosures only.

2/ IEC 60529 has Ingress Protection rating or International Protection (IP) markings. The first numeral marking is designation for protection of equipment against ingress of solid foreign objects (Dust Protection). Second numeral marking is designation for protection of equipment against ingress of water.

3/ NEMA 1, 2, 5, 7, 9, 12, 12K, and 13 are rated for Indoor use only.

4/ Class II, Group E, F, G Hazardous Locations per NFPA 70 designed to contain an internal explosion without causing an external hazard.

5/ Class I, Group A, B, C, D Hazardous Locations per NFPA 70 designed to contain an internal explosion without causing an external hazard.

6/ NEMA rated for temporary submersion.

7/ NEMA rated for prolonged submersion.

6.4 Subject term (key word) listing.

Airtight
Degree of enclosure
Dripproof
Dust-ignition proof
Dustproof
Dusttight
Environmental protection
Explosion-proof
Hermetic
Open-submersible
Splashproof
Spraytight
Submersible
Totally enclosed
Watertight

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – TE
Navy – SH

Preparing activity:
Navy – SH
(Project ENVR-2023-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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